

## Letter

# A step forward in fungal biomass estimation – a new protocol for more precise measurements of soil ergosterol with liquid chromatography-mass spectrometry and comparison of extraction methods

Due to the ecological and economical role of fungi, we need precise tools to estimate fungal biomass. A common method is to measure the fungal biomarker ergosterol with high-pressure liquid chromatography followed by UV detection. Here, we provide a comparison of extraction protocols, and we develop a new liquid chromatography-mass spectrometry method to improve the sensitivity of ergosterol measurements.

Boreal forests and forested peatlands store globally significant amounts of carbon (C) in the soil organic matter (SOM; Crowther *et al.*, 2016; Mäkipää *et al.*, 2023). However, climate change or ecosystem management may disturb a vulnerable balance between the rate of C accumulation and organic matter decomposition, thus shifting boreal forest C pools from sinks into C sources (Arneth *et al.*, 2017; Mäkipää *et al.*, 2023). Transformations of SOM are driven by fungi and bacteria in interaction with plant roots (Adamczyk *et al.*, 2019; Fanin *et al.*, 2022). Fungi are efficient decomposers of organic matter via secreted enzymes and also via non-enzymatic routes (Lindahl & Tunlid, 2015). Not only free-living fungi are powerful decomposers but also fungi living in symbioses with trees (ectomycorrhizal fungi) and with shrubs (ericoid mycorrhizal fungi; Kohler *et al.*, 2015; Lindahl *et al.*, 2021). Thus, precise measurements of fungal biomass in line with fungal activities are essential for accurately describing the C cycle of terrestrial ecosystems (Wallander *et al.*, 2013).

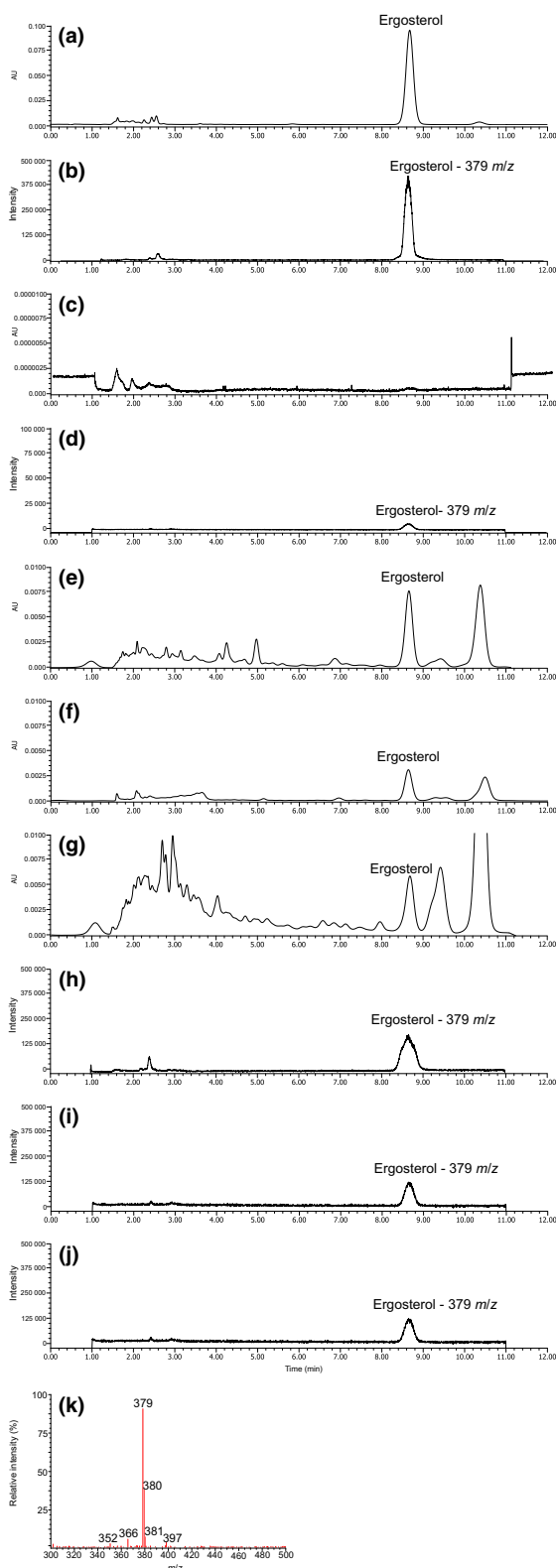
Estimation of fungal biomass is commonly based on a chemical marker, ergosterol (5,7-diene oxysterol), although alternative methods also exist (e.g. DNA and phospholipid fatty acids; Wallander *et al.*, 2013; Adamczyk *et al.*, 2020). Ergosterol is found exclusively in the membranes of living fungal cells, except for some microalgae (Newell *et al.*, 1987; Wallander *et al.*, 2013). Although ergosterol is widespread in more advanced fungal taxa as ascomycetes and basidiomycetes, some primitive fungi may contain sterols other than ergosterol (Olsson *et al.*, 2003). Moreover, ergosterol concentration is species-specific (Pasanen *et al.*, 1999). In addition, some studies have shown that ergosterol, under certain conditions,

can persist in dead mycelial tissues for considerable periods of time, leading to fungal biomass overestimation (Zhao *et al.*, 2005; Joergensen & Wichern, 2008). Despite these disadvantages, ergosterol analysis is considered a reliable tool, which is commonly used in laboratory practices.

There are numerous protocols for extracting ergosterol from different materials and usually the purity of the ergosterol peak in chromatograms is difficult to isolate and interpret due to possible co-elution with other compounds. Confirmation with mass spectrometry could improve the sensitivity/accuracy of the method. To fulfil the need to improve and take the next step in fungal biomass estimation, we provide an overview of the most common extraction protocols and compare their effectiveness. Moreover, we developed a liquid chromatography-mass spectrometry (LC-MS) method to ensure peak purity and detectability at low concentrations commonly seen in soils of low organic matter content.

Ergosterol extraction from soil involves extraction in organic solvent or saponification in alcoholic KOH, followed by incubation and liquid-liquid extraction (Gessner *et al.*, 1991). Most commonly, soil samples are extracted with methanol (e.g. Mille-Lindblom *et al.*, 2004), with 10% KOH in methanol or with 4% KOH in ethanol (Brodie *et al.*, 2003). Comparing these methods, we found that 10% KOH in methanol was the most effective extractant (Supporting Information Table S1). Saponification renders fungal cell membrane more permeable to methanol extraction, which explains why KOH in methanol is superior to methanol (Pastinen *et al.*, 2017). Second, liquid-liquid extraction is usually performed with one of the following solvents: pentane (Brodie *et al.*, 2003), hexane (Eash *et al.*, 1996) or cyclohexane (Frostegård & Bååth, 1996). The most effective solvent was cyclohexane, likely because of its high polarity index (for results see Tables S1, S2). We are aware that protocols differ in the incubation time and temperature; thus, we also used solvents from other protocols under the conditions of our protocol (i.e. pentane and hexane instead of cyclohexane and methanol instead of 10% KOH in methanol, for results see Table S2). The results strongly suggest that solvents are responsible for most of the differences in extraction yields (see Supporting Information for results and precise description of all used extraction protocols, Methods S1). Thus, we propose an extraction protocol based on Frostegård & Bååth (1996) as described in Adamczyk *et al.* (2019): 0.25 g (organic soil) or 0.5 g (mineral soil) of freeze-dried sample is extracted with 1 ml of cyclohexane and 4 ml of 10% KOH in methanol. After 15 min of ultrasonic bath, samples are incubated for 1 h in 70°C, and after cooling, 1 ml of water and 2 ml of cyclohexane are added. The tubes are vortexed for 1 min, and after centrifugation (2000 g, 5 min), the top phase is transferred to another test tube. The procedure with water and cyclohexane is repeated and the combined extracts are evaporated under N<sub>2</sub> gas at

40°C. The samples are re-dissolved in methanol by heating at 40°C for 15 min and filtered (0.2 µm PTFE filter). The amount of ergosterol is measured with HPLC using a C18 reverse-phase column (4.6 × 150 mm; Phenomenex, Torrance, CA, USA).



Ergosterol is analyzed using a 10 µl sample injection and isocratic separation with methanol (1 ml min<sup>-1</sup>) and detected at 282 nm (to be described later for novel LC-MS method).

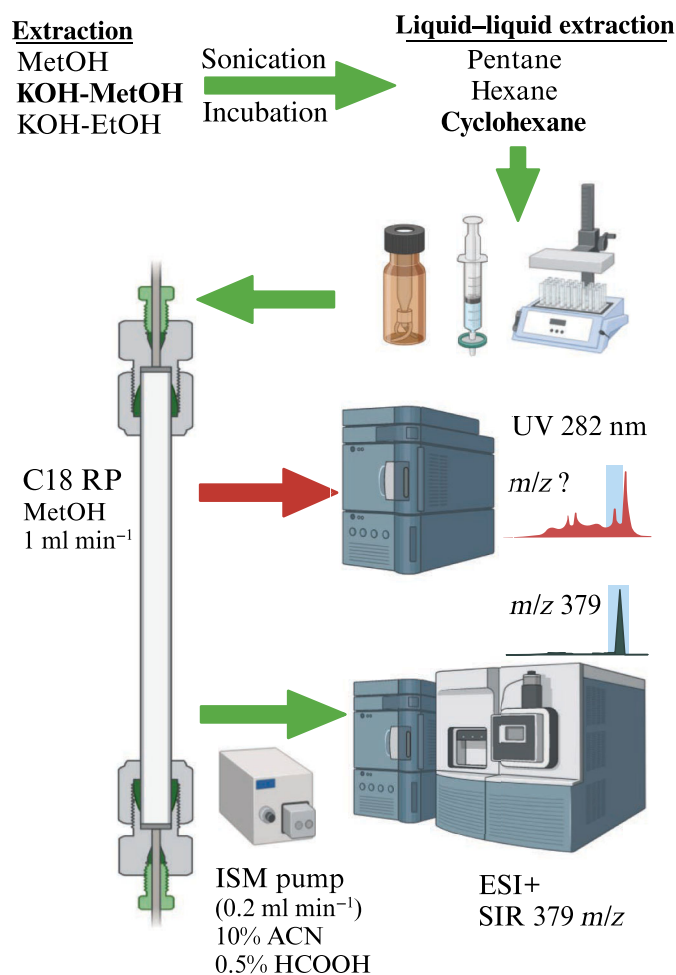
We developed a new LC-MS method to confirm the purity of ergosterol peak and a more precise method with mass spectrometry to quantify ergosterol concentration. Samples were run on an HPLC system (Arc HPLC; Waters) equipped with UV detector and mass spectrometer (Acquity qDa; Waters, Milford, MA, USA) with the EMPOWER 3.6 software. Similarly to the HPLC-UV method (Adamczyk *et al.*, 2019), a 10 µl sample prepared in line with the above-described protocol was injected into the HPLC. Separation was conducted using a C18 column (as described above) with the column effluent directed not only to the UV detector but also to the mass spectrometer with an isocratic solvent manager pump (ISM; Waters). A post-column solvent from the isocratic pump was used to improve ionization of the samples with no disturbance to the separation in the column. The post-column solvent consisted of 10% acetonitrile in water with 0.5% formic acid (0.2 ml min<sup>-1</sup>). Mass spectrometer (Acquity qDa; Waters) with electrospray ionization (ESI) in positive ion mode was used under the following conditions: cone voltage 10 V, probe temperature 600°C, and capillary voltage 0.8 kV.

The proposed LC-MS method has a limit of detection (for ergosterol standard) of 0.005 µg ml<sup>-1</sup> and a limit of quantitation of 0.0025 µg ml<sup>-1</sup> and for HPLC-UV these values are 10 times higher (0.05 µg ml<sup>-1</sup> and 0.025 µg ml<sup>-1</sup>, respectively). For a comparison of the HPLC-UV and LC-MS chromatograms, see Fig. 1a–j.

Although ionization with ESI usually results in little analyte degradation and in-source fragmentation, in some cases, ESI may lead to fragmentation (Ory *et al.*, 2020), making compound identification challenging. This was also the case for ergosterol. Because the molecular mass of ergosterol is 396 Da, in the positive ion mode, one could expect the main ion at *m/z* at 397 ([M+H]<sup>+</sup>). Instead, we observed a dominating ion at *m/z* 379 ([M-H<sub>2</sub>O+H]<sup>+</sup>) and also other ions, more precisely *m/z* 380 ([M-CH<sub>4</sub>+H]<sup>+</sup>), *m/z* 381 ([M-CH<sub>3</sub>+H]<sup>+</sup>), *m/z* 366 ([M-C<sub>2</sub>H<sub>6</sub>+H]<sup>+</sup>), and *m/z* 352 ([M-C<sub>3</sub>H<sub>8</sub>+H]<sup>+</sup>; Fig. 1k; For fragmentation pathways see Figs S1–S5). Thus, to obtain the highest precision, we propose to run mass spectrometry in selected ion recording (SIR) mode with an *m/z* of 379. For the scheme of the extraction protocol comparisons and development of the LC-MS method, see Fig. 2.

With the LC-MS method described above, we measured ergosterol concentration in organic and mineral soils from two forest sites (Karstula and Hyytiälä) as well as in a forested peat soil

**Fig. 1** Chromatograms of ergosterol standards and soil samples. (a) ergosterol standard (0.1 mg ml<sup>-1</sup>) in HPLC-UV, (b) ergosterol standard (0.1 mg ml<sup>-1</sup>) in liquid chromatography-mass spectrometry (LC-MS), (c) ergosterol standard (0.00001 mg ml<sup>-1</sup>) in HPLC-UV, (d) ergosterol standard (0.00001 mg ml<sup>-1</sup>) in LC-MS, (e) soil sample from the organic layer of the Karstula site, HPLC-UV, (f) soil sample from mineral layer of the Karstula site, HPLC-UV, (g) sample from peat of the Ränskälänkorpi site 10–20 cm depth, HPLC-UV, (h) soil sample from the organic layer of the Karstula site, LC-MS, (i) soil sample from the mineral layer of the Karstula site, LC-MS, (j) soil sample from peat of the Ränskälänkorpi site 10–20 cm depth, LC-MS, (k) example of ergosterol peak purity. Karstula and Hyytiälä sites are forests on mineral soil, and Ränskälänkorpi is a forested peatland.



**Fig. 2** Scheme of extraction protocol comparisons and development of liquid chromatography-mass spectrometry (LC-MS) method. Freeze-dried soil sample is at first extracted with KOH-methanol (KOH-MetOH) and sonicated for 15 min following incubation for 1 h at 70°C and liquid-liquid extracted with cyclohexane. Cyclohexane phase is evaporated to dryness, dry residue is dissolved in methanol (MetOH) and filtered into HPLC vials. Separation is conducted with C18 reversed phase (C18 RP) column eluted with 1 ml min<sup>-1</sup> MetOH. HPLC-UV method includes detection of the ergosterol concentration with UV (282 nm) and LC-MS method use electrospray ionization in positive ion mode (ESI+) and selected ion recording (SIR) to detect and quantify ergosterol. In the LC-MS method, post-column solvent from isocratic solvent manager (ISM) with acetonitrile (ACN) and formic acid (HCOOH) is used to improve ionization. Created with BioRender.com.

(Ränskälänkorpi; see Fig. 1 for chromatograms and Table 1 for results and Methods S2 for site description). Ergosterol estimates obtained with the new LC-MS protocol were slightly higher at all sites compared with the HPLC-UV method (see Table 1). Most importantly, the LC-MS method in SIR mode (*m/z* 379) yielded a clear chromatogram with no peaks close to the retention time of ergosterol, significantly increasing the reliability of the results.

In conclusion, significant differences between ergosterol protocols and their yield call for harmonization of methodologies. We propose an extraction protocol with KOH in methanol with cyclohexane for liquid-liquid extraction. The novel LC-MS method proved to be superior to the HPLC-UV method for soil samples because of the higher quality of the peaks and the

**Table 1** Concentration of ergosterol extracted with protocol<sup>1</sup> (see Supporting Information Methods S1) and measured with HPLC-UV and LC-MS.

| Site, location                           | Soil layer/<br>depth | Ergosterol<br>concentration,<br>µg g <sup>-1</sup> DW<br>HPLC-UV | Ergosterol<br>concentration,<br>µg g <sup>-1</sup> DW LC-MS |
|------------------------------------------|----------------------|------------------------------------------------------------------|-------------------------------------------------------------|
| Karstula (forest on<br>mineral soil)     | Organic layer        | 51.54 (11.4)                                                     | 52.1 (5.4)                                                  |
|                                          | Mineral layer        | 2.3 (0.7)                                                        | 2.5 (0.1)                                                   |
| Hyytiälä (forest<br>on mineral<br>soil)  | Organic layer        | 89 (15.2)                                                        | 90.2 (4.4)                                                  |
|                                          | Mineral layer        | 4 (1.2)                                                          | 5.1 (0.2)                                                   |
| Ränskälänkorpi<br>(forested<br>peatland) | 0–10 cm              | 92.4 (35.5)                                                      | 93.2 (4.1)                                                  |
|                                          | 10–20 cm             | 42.2 (17.5)                                                      | 42.5 (3.5)                                                  |
|                                          | 20–30 cm             | 27.2 (7.2)                                                       | 28.2 (2.2)                                                  |
|                                          | 30–40 cm             | 29.4 (9.5)                                                       | 30.1 (2.5)                                                  |
|                                          | 40–50 cm             | 30.4 (7.9)                                                       | 29.1 (2.9)                                                  |

For site descriptions see Methods S2. DW, dry weight; LC-MS, liquid chromatography-mass spectrometry. Results are averages of 4 extraction replicates with SD in brackets.

possibility of following peak purity. Moreover, the LC-MS method has 10 times lower detection limit than that of HPLC-UV.

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## Competing interests

None declared.

## Author contributions

SA and BA conceived the idea and conducted analyses. RM and AL contributed to the data and contributed to the data interpretation. SA led the manuscript writing, with contributions from all authors.

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## Data availability

The data that support the findings of this study are available in the Supporting Information of this article.

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## Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

**Fig. S1** Fragmentation pathway to *m/z* 379.

**Fig. S2** Fragmentation pathway to *m/z* 380.

**Fig. S3** Fragmentation pathway to *m/z* 381.

**Fig. S4** Fragmentation pathway to *m/z* 366.

**Fig. S5** Fragmentation pathway to *m/z* 352.

**Methods S1** Precise description of extraction protocols.

**Methods S2** Precise description of the study sites.

**Table S1** Concentration of ergosterol extracted with different protocols and measured with HPLC-UV.

**Table S2** Concentration of ergosterol measured with HPLC-UV after extraction protocol<sup>1</sup> with different solvents.

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**Key words:** ergosterol fragmentation, forest soil, mineral soil, peatlands, selected ion recording.

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